Lecture 8: Integration of Rational Functions by Partial Fractions

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Apply the long division to an improper rational function.
- ☐ Find the partial fraction decomposition of a proper rational function.
- ☐ Integrating rational functions by partial fractions.

Introduction

A rational function is a function of the form $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are two polynomials, with $g(x) \neq 0$.

A rational function is proper if $\deg(f) < \deg(g)$, and improper if $\deg(f) \geq \deg(g)$.

If the integrand is an improper function, we can first apply the long division to express it as the sum of a polynomial and a proper rational function.

To integrate a proper rational function, we express it as the sum of partial fractions (if possible).

General Description of the Method

Success in writing a rational function $\frac{f(x)}{g(x)}$ as a sum of partial fractions depends on three things:

- The degree of $f(x)$ must be less than the degree of $g(x)$. That is, the function must be proper. If isn't, divide $f(x)$ by $g(x)$ and work with the remainder term.
- We must know the factors of $g(x)$. In theory, any polynomial with real coefficients can be written as a product of real linear factors and real quadratic factors. In practice, the factors may be hard to find.
- The values of the undetermined coefficients form a system of n linear equations in n unknowns.

We say a quadratic polynomial is irreducible if it is not a product of two real linear factors. That is, it has no real root.

Partial Fraction Decompositions

1. Let $x - r$ be a linear factor of $g(x)$. Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. Then to this factor, assign the sum of the m partial fractions:

$$\frac{A_1}{(x-r)} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_m}{(x-r)^m}.$$

   Do this for each distinct linear factor of $g(x)$.

2. Let $x^2 + px + q$ be an irreducible quadratic factor of $g(x)$. In this case, $x^2 + px + q$ has no real roots. Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:

$$\frac{B_1 x + C_1}{(x^2 + px + q)} + \frac{B_2 x + C_2}{(x^2 + px + q)^2} + \cdots + \frac{B_n x + C_n}{(x^2 + px + q)^n}.$$

   Do this for each distinct quadratic factor of $g(x)$.

Partial Fraction Decompositions

3. Set the original fraction $f(x)/g(x)$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions.
4. Find the values of the undetermined coefficients.

Example: Find the partial fraction decomposition of $\frac{x^2-3x+8}{x^3-4x^2+4x}$.

Partial Fraction Decompositions

Find the partial fraction decomposition of $\frac{2x+1}{(x^2+x+1)^2}$.

Examples

$$\int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)}\, dx$$

Examples

$$\int \frac{6x+7}{(x+2)^2}\,dx$$

Examples

$$\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3}\, dx$$

Examples

$$\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx$$

Examples

$$\int \frac{dx}{x(x^2+1)^2}$$

Examples

$$\int \frac{4x^2 - 15x + 29}{(x-5)(x^2 - 4x + 13)}\, dx$$

Other Ways to Find the Coefficients

Using differentiation: Find A, B, and C in the equation

$$\frac{x-1}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}.$$

Other Ways to Find the Coefficients

Assigning numerical value to x: Find A, B, and C in the equation

$$\frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}.$$